

## Clarkson Institute Summer School: Examples Sheet 7 - Permutation groups

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### Symmetric and alternating groups

1. Calculate the inverse of the permutation  $(123)(45) \in S_5$ .

*Solution:* The first factor of this permutation sends  $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$ , so we just need to reverse the direction of the arrows. This gives  $1 \rightarrow 3 \rightarrow 2 \rightarrow 1$ , corresponding to the cycle  $(132)$ . Similarly, the second factor of this permutation sends  $4 \rightarrow 5 \rightarrow 4$ , which again reversing the direction of the arrows gives  $4 \rightarrow 5 \rightarrow 4$  (i.e. it stays the same). Thus the inverse is:

$$(132)(45).$$

2. Decompose the following permutations into: (i) their disjoint cycles; (ii) either an even or odd number of transpositions, as appropriate.

- (a)  $(12)(1234)(12)$ ;  
(b)  $(123)(1234)(132)$ ;  
(c)  $(123)(235)(345)(45)$ .

*Solution:*

- (a) (i)  $(1342)$ ; (ii)  $(1342) = (12)(14)(13)$ ;  
(b) (i)  $(123)(1234)(132) = (1423)$ ; (ii)  $(1423) = (13)(12)(14)$ ;  
(c) (i)  $(12)(345)$ ; (ii)  $(12)(345) = (12)(35)(34)$ .

3. Suppose that a permutation has a disjoint cycle representation comprising cycles of lengths  $n_1, n_2, \dots, n_k$ . Show that the order of the permutation is the least common multiple of  $n_1, n_2, \dots, n_k$ . Hence, find the orders of the permutations in Question 1.

*Solution:* Let  $\sigma = \sigma_1 \dots \sigma_k$  be the disjoint cycle representation, with  $\sigma_i$  of length  $n_i$ . Then:

$$\sigma^r = \sigma_1^r \dots \sigma_k^r,$$

since disjoint cycles commute. This is equal to the identity if and only if  $\sigma_i^r = \text{id}$  for each  $i$ , since the cycles are disjoint. The smallest time this occurs is the least integer which contains  $n_1, n_2, \dots, n_k$  all as factors. This is the least common multiple as required.

The orders of the permutations are: (a) 4; (b) 4; (c) 6.

4. Let  $\sigma : \mathbb{Z}_{31} \rightarrow \mathbb{Z}_{31}$  be defined by  $\sigma(x) = 2x$ . Show that  $\sigma$  is a permutation. Reduce  $\sigma$  to its disjoint cycle representation, and find its order.

*Solution:* The map  $\sigma : \mathbb{Z}_{31} \rightarrow \mathbb{Z}_{31}$  given by  $\sigma(x) = 2x$  evidently has the inverse  $\sigma^{-1}(x) = 2^{-1}x = 16x$ , since  $2^{-1} = 16$  modulo 31 (in the multiplicative group). Thus it is a permutation. We have:

$$\sigma(1) = 2, \quad \sigma(2) = 4, \quad \sigma(4) = 8, \quad \sigma(8) = 16, \quad \sigma(16) = 1.$$

Hence one of the cycles is  $(1\ 2\ 4\ 8\ 16)$ . Next, we have:

$$\sigma(3) = 6, \quad \sigma(6) = 12, \quad \sigma(12) = 24, \quad \sigma(24) = 17, \quad \sigma(17) = 3.$$

Hence another of the cycles is  $(3\ 6\ 12\ 24\ 17)$ . Next, we have:

$$\sigma(5) = 10, \quad \sigma(10) = 20, \quad \sigma(20) = 9, \quad \sigma(9) = 18, \quad \sigma(18) = 5.$$

Hence another of the cycles is  $(5\ 10\ 20\ 9\ 18)$ . Next, we have:

$$\sigma(7) = 14, \quad \sigma(14) = 28, \quad \sigma(28) = 25, \quad \sigma(25) = 19, \quad \sigma(19) = 7.$$

Hence another of the cycles is  $(7\ 14\ 28\ 25\ 19)$ . Next, we have:

$$\sigma(11) = 22, \quad \sigma(22) = 13, \quad \sigma(13) = 26, \quad \sigma(26) = 21, \quad \sigma(21) = 11.$$

Hence another of the cycles is  $(11\ 22\ 13\ 26\ 21)$ . Finally, we have:

$$\sigma(15) = 30, \quad \sigma(30) = 29, \quad \sigma(29) = 27, \quad \sigma(27) = 23, \quad \sigma(23) = 15.$$

Hence the final cycle is  $(15\ 30\ 29\ 27\ 23)$ . Note also that  $\sigma(0) = 0$ . Hence the disjoint cycle representation is:

$$\sigma = (1\ 2\ 4\ 8\ 16)(3\ 6\ 12\ 24\ 17)(5\ 10\ 20\ 9\ 18)(7\ 14\ 28\ 25\ 19)(11\ 22\ 13\ 26\ 21)(15\ 30\ 29\ 27\ 23)(0).$$

Overall then, the order of  $\sigma$  is 5, since all the cycles have length 5 (except  $(0)$ ).

5. Find the largest possible orders of elements in  $S_5$  and  $A_5$ , respectively.

*Solution:* In  $S_5$ , we seek integers  $n_1, n_2, \dots, n_k$  satisfying  $n_1 + \dots + n_k = 5$  which maximise the least common multiple of the integers. The largest the integers can be is 5; this clearly does not give the largest least common multiple. The next largest is 4, but this would only leave 1, and hence cannot maximise the least common multiple. The next largest is 3, which when combined with 2 gives the least common multiple 6. This is the largest the least common multiple can become, so the greatest order is 6.

In  $A_5$ , the cycle type with length two and length three, e.g.  $(12)(345) = (12)(35)(34)$  cannot be used because it is an odd permutation. Therefore the best we can do is a cycle of type  $(12345) = (15)(14)(13)(12)$ , which has greatest order 5.

6. Show that the symmetric group  $S_n$  is generated by each of the following subsets:

- (a)  $\{(j, j+1) : 1 \leq j < n\}$ ;
- (b)  $\{(1, k) : 1 < k \leq n\}$ ;
- (c)  $\{(12), (12\dots n)\}$ .

*Solution:*

- (a) We know that any permutation is generated by transpositions. Observe that for a transposition  $(ab)$  with  $a < b$ , we can build this up from transpositions of the form  $(j, j+1)$  by doing the following:

$$(ab) = (a \ a+1)(a+1 \ a+2)\dots(b-1 \ b)(b-2 \ b-1)\dots(a+1 \ a+2)(a \ a+1).$$

Hence, the set generates.

- (b) Observe that any transposition  $(ab)$  can be written as  $(ab) = (1a)(1b)(1a)$ . Hence the set generates.

- (c) Here, we use conjugation. Define  $\sigma = (12\dots n)$ . Then:

$$\sigma(12)\sigma^{-1} = (23),$$

using the result on conjugation from the lectures. Similarly,  $\sigma^2(12)\sigma^{-2} = (34)$ , etc. Thus we can generate everything we have in part (a), and thus this set generates.

### Cayley's theorem

7. Prove the following generalisation of Cayley's theorem: if  $H \leq G$  is a subgroup of a group  $G$ , then there exists a normal subgroup  $K \trianglelefteq G$  with  $K \leq H$  such that  $G/K$  is isomorphic to a subgroup of  $\text{Sym}(G/H)$ . [Here,  $G/H$  denotes the set of left cosets of  $H$  in  $G$ , which is not a group unless  $H$  is normal. Note that Cayley's theorem is the special case  $H = \{e\}$ .]

*Solution:* Define a map  $\alpha_g : G/H \rightarrow G/H$  which transforms the cosets via  $\alpha_g(g'H) = (gg')H$ . This is well-defined, since if  $g'H = g''H$ , we have  $g'' = g'h$  for some  $h \in H$ . This gives  $\alpha_g(g''H) = (gg'')H = (gg'h)H = (gg')H$ . It is also invertible with inverse  $\alpha_{g^{-1}}$ , so is a permutation of  $G/H$ .

We now define a map  $f : G \rightarrow \text{Sym}(G/H)$  via:

$$f(g) = \alpha_g.$$

We claim that  $f$  is a homomorphism. Indeed, we note that:

$$\alpha_{gg'}(g''H) = (gg'g'')H = \alpha_g((g'g'')H) = \alpha_g(\alpha_{g'}(g''H)) = (\alpha_g \circ \alpha_{g'})(g''H).$$

Thus  $\alpha_{gg'} = \alpha_g \circ \alpha_{g'}$ . As a result, we have:

$$f(gg') = \alpha_{gg'} = \alpha_g \circ \alpha_{g'} = f(g)f(g'),$$

so indeed this is a homomorphism. Thus by the isomorphism theorem, we have  $G/\text{Ker}(f) \cong \text{Im}(f) \leq \text{Sym}(G/H)$ . So in particular taking  $K = \text{Ker}(f)$ , we have a normal subgroup  $K \trianglelefteq G$  such that  $G/K$  is isomorphic to a subgroup of  $\text{Sym}(G/H)$ .

8. Show that every finite group is isomorphic to a subgroup of an alternating group.

*Solution:* Cayley's theorem tells us that every finite group is isomorphic to a subgroup of some symmetric group  $S_n$ , hence it is sufficient to embed  $S_n$  into an alternating group. The idea is to extend any odd permutation in  $S_n$  with some extra transposition that doesn't affect the permutation, by embedding  $S_n$  into  $A_{n+2}$ . Define a map  $f : S_n \rightarrow A_{n+2}$  via:

$$f(\sigma) = \begin{cases} \sigma, & \text{if } \sigma \in A_n; \\ \sigma(n+1 \ n+2), & \text{if } \sigma \notin A_n. \end{cases}$$

We claim that this is a homomorphism. Splitting into cases, we have:

$$\begin{aligned} f(\sigma_1)f(\sigma_2) &= \begin{cases} \sigma_1\sigma_2, & \text{if } \sigma_1, \sigma_2 \in A_n, \\ \sigma_1(n+1 \ n+2)\sigma_2, & \text{if } \sigma_1 \notin A_n, \sigma_2 \in A_n, \\ \sigma_1\sigma_2(n+1 \ n+2), & \text{if } \sigma_1 \in A_n, \sigma_2 \notin A_n, \\ \sigma_1(n+1 \ n+2)\sigma_2(n+1 \ n+2), & \text{if } \sigma_1, \sigma_2 \notin A_n. \end{cases} \\ &= \begin{cases} \sigma_1\sigma_2, & \text{if } \sigma_1, \sigma_2 \in A_n \text{ or both } \sigma_1, \sigma_2 \notin A_n, \\ \sigma_1\sigma_2(n+1 \ n+2), & \text{if one of } \sigma_1, \sigma_2 \text{ is in } A_n, \text{ and the other is not.} \end{cases} \\ &= f(\sigma_1\sigma_2). \end{aligned}$$

Next, observe that the kernel is clearly the identity permutation only. Hence by the isomorphism theorem, we have  $S_n \cong S_n/\{e\} \cong \text{Im}(f) \leq A_{n+2}$ . Thus  $S_n$  is isomorphic to a subgroup of an alternating group, so every finite group is isomorphic to a subgroup of an alternating group.

9. Prove that  $S_n$  has a subgroup isomorphic to  $Q_8$  if and only if  $n \geq 8$ . [Hint: consider  $H \subseteq Q_8$ , the set of all elements whose image permutations fix 1. Begin by proving this is a subgroup, and consider its cosets.]

*Solution:* We have  $Q_8$  isomorphic to a subgroup of  $S_8$  by Cayley's theorem. Certainly for  $S_n$ , we have that  $S_8 \leq S_n$ , so again  $Q_8$  can be viewed as isomorphic to a subgroup of  $S_n$  for  $n \geq 8$ .

For  $n \leq 7$ , suppose for a contradiction that there exists an injective homomorphism  $f : Q_8 \rightarrow S_n$  where  $n \leq 7$ . Let  $H \subseteq Q_8$  be the set of all elements whose image permutations fix 1, that is:

$$H = \{g \in Q_8 : f(g)(1) = 1\}.$$

Then  $H$  is a subgroup, since:

- (G1) *Closure.* If  $x, y \in H$ , then  $f(x)(1) = 1$  and  $f(y)(1) = 1$ . Therefore  $f(xy)(1) = f(x)(f(y)(1)) = f(x)(1) = 1$ . So  $xy \in H$ .
- (G2) *Associativity.* Inherited from  $Q_8$ .
- (G3) *Identity.*  $f(e) = \text{id}$ , so fixes 1.
- (G4) *Inverses.* If  $f(g)$  is a permutation sending  $1 \rightarrow 1$ , then  $f(g)^{-1} = f(g^{-1})$  is a permutation also sending  $1 \rightarrow 1$ .

Next, consider a  $xH$  of  $H$ . Certainly if  $x \notin H$ , then this permutes 1, giving  $f(xh)(1) = f(x)(f(h)(1)) = f(x)(1) \neq 1$  by definition, if  $h \in H$ . Further, we claim that distinct cosets permute 1 differently. For suppose that  $y, z \in xH$ . Then we have  $y = xh, z = xh'$  for some  $h, h' \in H$ . This gives:

$$f(y)(1) = f(x)(f(h)(1)) = f(x)(1), \quad f(z)(1) = f(x)(f(h')(1)) = f(x)(1),$$

so all elements of the coset map 1 to the same location in  $\{1, 2, \dots, n\}$ . Further, if there exist elements  $x, y \in Q_8$  such that  $f(x)(1) = f(y)(1)$ , then we have  $f(x^{-1}y)(1) = 1$  by the homomorphism property, giving  $x^{-1}y \in H$ , and hence  $y \in xH$ . Thus  $x, y$  belong to the same coset of  $H$ .

It follows that there are at most 7 cosets of  $H$  inside  $Q_8$ , since there are only 7 distinct locations to which 1 can be sent. If  $m$  is the number of cosets then, we have:

$$m|H| = 8 \quad \Rightarrow \quad |H| > 1.$$

In particular, we see that  $H$  must contain at least two elements. But note that  $-1$  is in every subgroup of  $Q_8$ , so  $-1 \in H$ .

The argument did not depend on the element 1 which we were fixing. Therefore, the same applies to the subgroup fixing 2, the subgroup fixing 3, etc. Therefore  $-1$  is in each of these subgroups, and thus  $f(-1) = \text{id}$ . This implies that  $f$  is not injective, as required.  $\square$

### Conjugacy in permutation groups

10. By considering the possible cycle types, determine the sizes of all the conjugacy classes in both  $S_4$  and  $S_5$ .

*Solution:* In  $S_4$ , the cycle types have representatives  $(1234)$ ,  $(123)$ ,  $(12)$ ,  $(12)(34)$ . There is also the identity, which is its own conjugacy class. Therefore the complete collection of conjugacy class sizes is:

$$|\text{Cl}((1234))| = \frac{4!}{4} = 6, \quad |\text{Cl}((123))| = \frac{4 \cdot 3 \cdot 2}{3} = 8, \quad |\text{Cl}((12))| = \frac{4 \cdot 3}{2} = 6,$$

$$|\text{Cl}((12)(34))| = \frac{4 \cdot 3 \cdot 2 \cdot 1}{2 \cdot 2 \cdot 2} = 3, \quad |\text{Cl}(\text{id})| = 1.$$

This gives a total of  $6 + 8 + 6 + 3 = 24$  elements, which is expected, as  $|S_4| = 4! = 24$ .

In  $S_5$ , the cycle types has representatives  $(12345)$ ,  $(1234)$ ,  $(123)$ ,  $(123)(45)$ ,  $(12)$ ,  $(12)(34)$ , and the identity, which is its own conjugacy class. Therefore the complete collection of conjugacy class sizes is:

$$|\text{Cl}((12345))| = \frac{5!}{5} = 24, \quad |\text{Cl}((1234))| = \frac{5!}{4} = 30, \quad |\text{Cl}((123))| = \frac{5 \cdot 4 \cdot 3}{3} = 20,$$

$$|\text{Cl}((123)(45))| = \frac{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{3 \cdot 2} = 20, \quad |\text{Cl}((12))| = \frac{5 \cdot 4}{2} = 10,$$

$$|\text{Cl}((12)(34))| = \frac{5 \cdot 4 \cdot 3 \cdot 2}{2 \cdot 2 \cdot 2} = 15, \quad |\text{Cl}(\text{id})| = 1.$$

This gives a total of  $24 + 30 + 20 + 20 + 10 + 15 + 1 = 120$ , which is expected, as  $|S_5| = 5! = 120$ .

11. (a) By considering the chiral cubic group  $O$ , give a geometrical interpretation of the conjugacy classes in  $S_4$ .
- (b) By considering the chiral tetrahedral group  $T$ , give a geometrical interpretation of the conjugacy classes in  $A_4$ .

*Solution:*

- (a) There are four diagonals of the cube, and any rotation of the cube permutes these diagonals - we can do so arbitrarily with rotations. This implies that the chiral cubic group is isomorphic to  $S_4$ . The conjugacy classes can be split as:
- The identity, which does nothing.
  - The transpositions, which are rotations by  $180^\circ$  about an axis through the midpoints of two opposite edges, which passes through the centre.
  - The three-cycles, which are  $120^\circ$  and  $180^\circ$  rotations about one of the diagonal axes.
  - The double transpositions, which are rotations by  $\pi$  about an axis through the centres of opposite faces.
  - The four-cycles, which are rotations by  $\pi/2$  and  $3\pi/2$  about an axis through the centres of opposite faces.
- (b) We know that the chiral tetrahedral group  $T$  has size 12, and is a subgroup of the full tetrahedral group  $T_h \cong S_4$ . Since every permutation preserves the handedness of the tetrahedron, it must be that the induced permutation of the vertices is an even permutation. As a result, we have  $T \cong A_4$ . The conjugacy classes can be split as:
- The identity, which does nothing.
  - The double transpositions, which are rotations by  $180^\circ$  about an axis through the midpoints of opposite edges.
  - The three-cycles fall into two classes: the  $120^\circ$  rotations and the  $240^\circ$  rotations, which cannot be conjugated into one another using rotations but not reflections.

12. Show that  $A_5$  is not a simple group. [Hint: start by calculating the sizes of the conjugacy classes in  $A_5$ .]

*Solution:* Representatives of each of the possible cycle types in  $A_5$  are  $(12345)$ ,  $(123)$ ,  $(12)(34)$ , and the identity. Examining each of the cycle types in turn:

- For the identity, the conjugacy class size is 1. Clearly all odd permutations commute with the identity, and hence it does not split (it also has odd size, so cannot split).
- For  $(12)(34)$ , the conjugacy class size in  $S_5$  is  $(5 \cdot 4 \cdot 3 \cdot 2) / (2 \cdot 2 \cdot 2) = 15$ . Note that this permutation obviously commutes with  $(12)$ , and hence the conjugacy class does not split (again, it also has odd size, so cannot split).
- For  $(123)$ , the conjugacy class size in  $S_5$  is  $(5 \cdot 4 \cdot 3) / 3 = 20$ . This permutation obviously commutes with  $(45)$ , and hence the conjugacy class does not split.
- Finally for  $(12345)$ , the conjugacy class size in  $S_5$  is  $5!/5 = 24$ . But then the centraliser of  $(12345)$  in  $S_5$  must have size 5, so it cannot split, thus the conjugacy class must split. Therefore the conjugacy class splits into two conjugacy classes of size 12 each.

Therefore the conjugacy classes, whose sizes must be summed to give the size of any candidate normal subgroup (which is the union of conjugacy classes), have sizes 1, 15, 20, 12, 12. We now

hunt for normal subgroups. The size of  $A_5$  is 60, so by Lagrange the possible normal subgroup sizes are:

- 1: can only be expressed as the identity conjugacy class  $\{\text{id}\}$ ; trivially normal.
- 2: is not the sum of conjugacy class sizes.
- 3: is not the sum of conjugacy class sizes.
- 4: is not the sum of conjugacy class sizes.
- 5: is not the sum of conjugacy class sizes.
- 6: is not the sum of conjugacy class sizes.
- 10: is not the sum of conjugacy class sizes.
- 12: is not the sum of conjugacy class sizes.
- 15: is not the sum of conjugacy class sizes.
- 20: is not the sum of conjugacy class sizes.
- 30: noting that we must include the conjugacy class 1, if we include 20, we cannot make up the size. Similarly if we include 15 we cannot make up the size. Including both 12's cannot make up the size either.

So it is impossible to build a proper non-trivial normal subgroup from the given conjugacy class sizes. Hence  $A_5$  is simple.